École Nationale des Ponts et Chaussées

Département de première année

Bases scientifiques pour la transition énergétique 1

Séance 7 : Récapitulatif des points essentiels partie fluide

2026

Recap of what must be known for the exam

1. Basics of fluid mechanics

At the continuum scale in the Eulerian description, matter at position $\mathbf{x}$ and time t is described by its local intensive state parameters: *mass density* $\rho(\mathbf{x},t)$ and *internal energy density* $\rho(\mathbf{x},t)\,u(\mathbf{x},t)$. According to thermodynamics, these state parameters are conjugated to the *stress tensor* $\underline{\underline{\sigma}}(\mathbf{x},t)$ and *temperature* $T(\mathbf{x},t)$, respectively.

The macroscopic motion of matter is described by the *velocity* $\mathbf{v}(\mathbf{x},t)$. Therefore, the macroscopic *kinetic energy density* is given by $\rho(\mathbf{x},t)\,e_{kin}(\mathbf{x},t) = \frac{1}{2}\rho(\mathbf{x},t)\,|\mathbf{v}(\mathbf{x},t)|^2$.

The *kinetic energy flux* is given by $\mathbf{J}_{e_{kin}}(\mathbf{x},t) = \rho(\mathbf{x},t)\,e_{kin}(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t)$. The magnitude of this flux $\mathcal{P} = |\mathbf{J}_{e_{kin}}(\mathbf{x},t)| = \frac{1}{2}\rho(\mathbf{x},t)\,|\mathbf{v}(\mathbf{x},t)|^3$ represents the *power density* of the flow of matter that one tries to recover with a wind turbine. The power density of a 30 km/h wind is $\mathcal{P} = 1\,740$ W/m^2 is approximately equivalent to that of a 5 km/h water stream, and this amount is of similar magnitude as the solar radiation before entering the atmosphere ($1\,380$ W/m^2).

The time evolution of a continuum system is governed by conservation laws. The conservation of a quantity $A(\mathbf{x},t)$ can be written both in an integral form, and in a local form :

$$\int_\Omega \frac{\partial A}{\partial t} dV + \int_{\partial\Omega} \mathbf{J}_A.\mathbf{n} dA = \int_{\partial\Omega} \mathbf{q}_A.\mathbf{n} dA + \int_\Omega S_A dV$$

$$\frac{\partial A}{\partial t} + \nabla.\mathbf{J}_A = \nabla.\mathbf{q}_A + S_A$$

where $\mathbf{J}_A = A(\mathbf{x},t)\,\mathbf{v}$ is the convective flux of A, $\mathbf{q}_A(\mathbf{x},t)$ is the non-convective flux of A, and $S_A(\mathbf{x},t)$ is the external volumetric source of A.

In fluid mechanics, the time evolution of a continuum system is governed by three conservation laws, presented hereafter in their local form:

Conservation of mass also called the *continuity equation*:

$$\frac{\partial \rho}{\partial t} + \nabla.\mathbf{J}_\rho = 0 \text{ with } \mathbf{J}_\rho(\mathbf{x},t) = \rho(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t) \text{ the mass flux}$$

Conservation of linear momentum also called the *Navier-Stokes equation*:

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}.\nabla\mathbf{v} = \frac{\nabla.\underline{\underline{\sigma}}}{\rho} + \mathbf{g}$$

where $\mathbf{g}$ is the acceleration due to gravity.

Conservation of energy also called the *energy balance*:

$$\frac{\partial(\rho e)}{\partial t} + \nabla.(\mathbf{J}_e) = \nabla.\left(\underline{\underline{\sigma}}.\mathbf{v}\right) + \rho\mathbf{g}.\mathbf{v} + \nabla.\mathbf{q} \text{ with } \mathbf{J}_e(\mathbf{x},t) = \rho(\mathbf{x},t)\, e(\mathbf{x},t)\, \mathbf{v}(\mathbf{x},t) \text{ the energy flux}$$

where $\mathbf{q}$ is the heat flux.

There is also a conservation of the angular momentum, which requires the stress tensor $\underline{\underline{\sigma}}$ to be symmetrical. The three conservation laws correspond to a total of 5 equations (linear momentum is vectorial), and there are five unknowns $(\rho, T, \mathbf{v})$. To fully set the problem, one must provide a behavior law relating the internal energy u and stress tensor $\underline{\underline{\sigma}}$ to the unknowns (e.g., ideal gas law) and a heat flux law relating $\mathbf{q}$ to the unknowns (e.g., Fourier's law).

For air and water flows, one can consider a few simplifying assumptions: the fluid is *incompressible*, its evolution is *adiabatic*, and its behavior is *Newtonian* (in the spherical and deviatoric decomposition of the stress tensor $\underline{\underline{\sigma}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$, the deviatoric part is given by $\underline{\underline{\tau}} = \mu\left(\nabla\mathbf{v} + \nabla\mathbf{v}^T\right)$ with μ the dynamic viscosity). Under these simplifying assumptions, the conservation laws become:

$$\begin{cases} \nabla.\mathbf{v} = 0 & \text{continuity equation} \\ \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}.\nabla\mathbf{v} = -\frac{\nabla p}{\rho_0} + \frac{\mu}{\rho_0}\Delta\mathbf{v} + \mathbf{g} & \text{Navier-Stokes equation} \\ \frac{\partial e}{\partial t} + \mathbf{v}.\nabla e = -\frac{\mathbf{v}.\nabla p}{\rho_0} + \frac{\nabla.\left(\underline{\underline{\tau}}.\mathbf{v}\right)}{\rho_0} + \mathbf{g}.\mathbf{v} & \text{energy balance} \end{cases}$$

Additionally, sometimes we can assume a stationary evolution ($\frac{\partial X}{\partial t} = 0$), and an ideal fluid (non-viscous fluid, $\mu = 0$).

Another formulation of the conservation of energy is given by *Bernoulli's principle*. For a stationary flow of an incompressible ideal fluid,

$$\text{the quantity } \frac{\mathbf{v}^2}{2} + \frac{p}{\rho_0} + gz \text{ is conserved along the flow}$$

Thus, the pressure of a fluid drops as its velocity (or its altitude) increases. The quantity $p_{dyn} = \frac{1}{2}\rho_0\mathbf{v}^2$ is called the *dynamical pressure* and represents the maximum increase of pressure as the fluid velocity is reduced to zero.

2. Working principle of an airfoil

For a closed curve $\mathcal{C}$, the line integral $\Gamma(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{v}.d\mathbf{x}$ is called the *circulation*. Applying the Stokes theorem, the circulation can be expressed as the flow of $\boldsymbol{\omega} = \nabla \times \mathbf{v}$, called *vorticity*, through the section delimited by $\mathcal{C}$: $\Gamma(\mathcal{C}) = \int_S \boldsymbol{\omega}.d\mathbf{S}$. Circulation and vorticity characterize the global and local rotation of a fluid. A fluid with no vorticity is said irrotational. For an ideal fluid, the vorticity of a packet of fluid is constant as the packet moves through the flow (Kelvin's circulation theorem). In particular, an ideal fluid that is irrotational will remain irrotational.

The example of a ball moving through the air (Figure **??**) shows that a lift force appears when the ball is rotating, because, by application of Bernouilli's principle, the circulation of the air velocity around the ball favors lower air pressures on one side and higher air pressures on the other side. The *Kutta-Zhukovskii theorem* provides a formal demonstration of the relationship between lift and circulation: a stationary flow of an incompressible ideal fluid

around an object exerts no drag and a lift proportional to the circulation of the fluid around the object:

$$\begin{cases} f_{\text{lift}} = \rho_0 v_\infty \Gamma(\mathcal{C}) \\ f_{\text{drag}} = 0 \end{cases}$$

where v_∞ is the free-stream velocity of the fluid, and $\mathcal{C}$ is any closed curve around the object oriented counterclockwise. The fact that there is no drag is known as d'*Alembert's paradox*. The absence of drag originates from the absence of viscosity of the fluid. Real fluids have non-zero viscosity and there is always a non-zero drag force.

An airfoil is designed to maximize lift while avoiding drag. The shape of the airfoil (in particular the sharp trailing edge) and its *angle of attack* α, induce a non zero circulation of the air around it which causes the lift. Yet, α must be kept small otherwise a region of turbulence develops behind the airfoil and the lift force drops.

By dimensional analysis, one can show that, for a stationary flow around around an object, the lift and drag force are of the form:

$$\begin{cases} f_{drag} = c_d(\alpha) \frac{\rho_0 v_\infty^2}{2} K \\ f_{\text{lift}} = c_l(\alpha) \frac{\rho_0 v_\infty^2}{2} K \end{cases}$$

where v_∞ is the free flow velocity, K is the chord length of the airfoil, and $c_d(\alpha)$ and $c_l(\alpha)$ are dimensionless coefficients called the *drag and lift coefficients*. For a flow a fixed Reynolds number and a object of fixed reduced shape (i.e., same shape upon homothetic expansion or contraction), the drag and lift coefficients depend only on the angle of attack α. Their precise estimation requires computational fluid mechanics or wind tunnel experimental tests. An example of airfoil used for wind energy is given in Fig. **??**. Optimal lift versus drag force ratio is achieved for angles of attack around 5°-10°.

3. Application to energy

A wind turbine (or a hydrofoil) capturing the kinetic energy from a free flow of air (or water) cannot capture all the kinetic power of the flow. Indeed, as the fluid passes through the turbine, it must keep some of its velocity to continue moving otherwise all the flow would bypass the turbine. The maximum efficiency, known as the *Betz limit*, is $\frac{16}{27} \approx 59.3\%$. It is derived from the conservation of mass, energy and axial momentum applied to the fluid flow passing through the turbine, and it holds irrespective of the nature of the turbine. In the particular case of horizontal axis wind turbines, the Betz limit can be further refined by taking into account the rotation of the fluid in the wake of the turbine (Fig. **??**).

A horizontal axis wind turbine is based on airfoils that are designed to convert a maximum of the fluid kinetic energy into rotational power. As the blades are rotating, the apparent velocity differs from the center of the turbine to the tips of the blades. The orientation and chord length are varying along the blades to reach optimum efficiency for a given *tip-speed ratio* (rotation speed at the tip of a blade divided by the far field velocity of the incoming fluid). Also, one usually considers 3 blades, which is the minimum number of blades with no inertia effects. But sometimes higher number of blades is used, for instance to favor torque over power at low rotation velocities.

Hydropower is not limited by the Betz limit and can reach ideal efficiency, because the flow of the fluid is constrained to go through the turbine and no by-pass is possible. Hydropower converts the potential energy of stored water ($\rho g z$) into mechanical energy.